CO-1 - Tool-1

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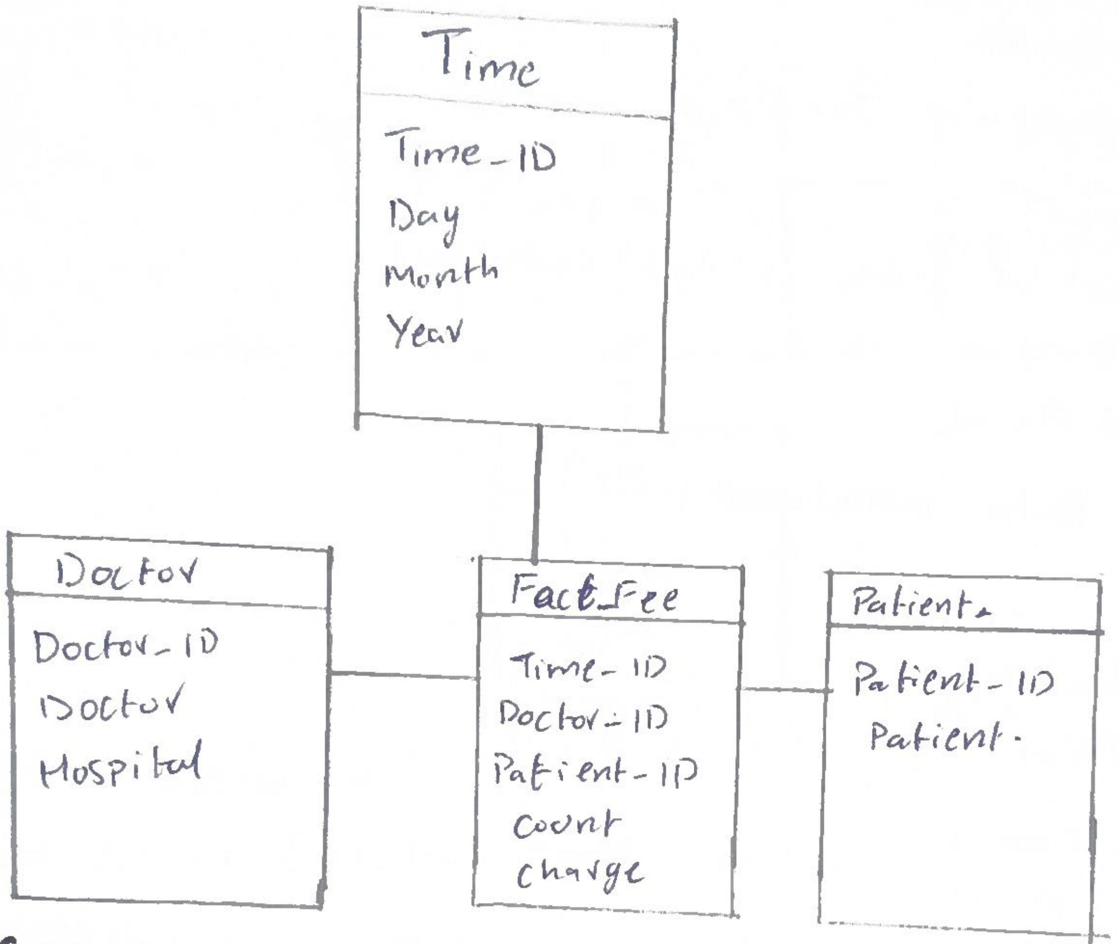
1) a) Three Popular Schemas used for data warehouses
* Star Schema
* Snowflake Schema
* Fact Constellation (Galaxy) Schema

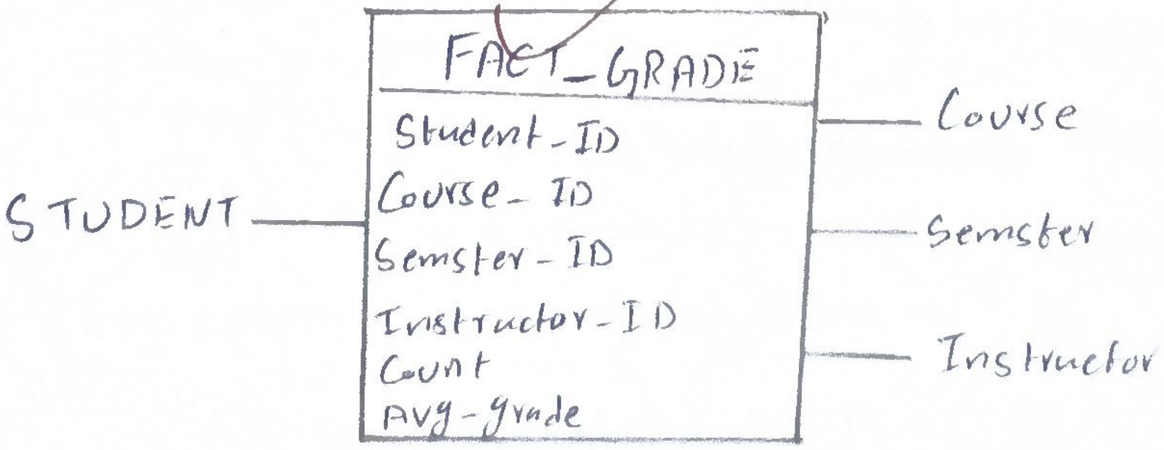
b) Star Schema Diagram.



c) SQL Query.

```
SELECT year, month, doctor, Patient,
       SUM(count) As Total_Visits,
       SUM(charge) As Total_charge
From Fee
Group By year, month, doctor, Patient;
```

2) a) Snowflake Schema.



b) OLAP Operations.

Starting Cuboid:

[Student, Course, Semster, Instructor]

Operations:

* Roll UP Semster -> Year
* Slice Course = CS
* Aggregate Avg-Grade
* Display average grade for each Student.

c) Number of Cubiods.

Formula = Number of levels in each dimensions

$= 5 \times 5 \times 5 \times 5$

$= 625$ Cubiods

3) Weather Bureau Data Warehouse

Dimensions :- → Time
→ Location
→ Probe.

Measure :- → Temperature
→ Air Pressure
→ Precipitation

Advantages :- → Supports OLAP analysis
→ Identifies weather trends
→ Fast multidimensional queries

4) Data Cube Problem

Given: * n dimensions
* P values per dimension.

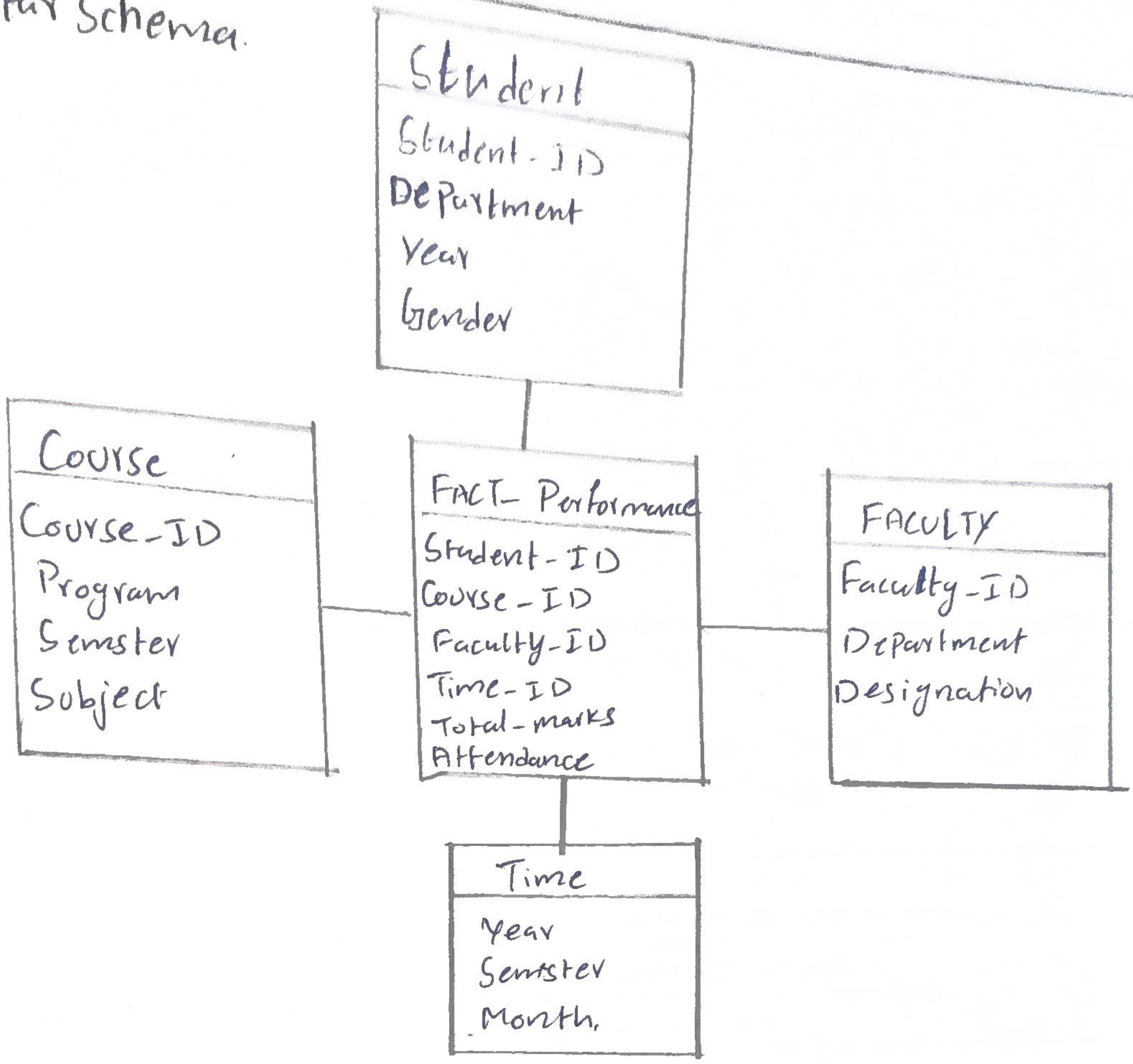
a) Maximum Cells in base Cuboid :- P^n

b) Minimum Cells in base Cuboid :- 1

c) Maximum Cells in entire Cube :- $(P+1)^n$

d) Minimum Cells in entire Cube :- 2^n

5) a) Star Schema.



b) OLAP Operations.

Base Cubiod:- [Student, Course, Faculty, Time].

Operations:- * Roll up Student → Department

* Roll up Time → Academic Year

* Aggregate Average Marks

Result:- Department-wise average marks for each academic year.

c) Number of Cubiods.

Levels = students = 4

Course = 3

Faculty = 3

Time = 5

Formula:- 4 × 3 × 3 × 5 = 180.

= 180 Cubiods //